

CloudWalk Payments Inc.

Responsible Use of Large Language Models

in BSA/AML Compliance Workflows

Technical Proposal — Workflow 1: SAR Narrative Auto-Draft · Workflow 2: Tier-1 Alert Triage Summarization

Prepared for: ** **BSA/AML Compliance Committee · Sponsor Bank (Lead Bank, FDIC member) · Regulatory Examiner Review

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Classification: ** **Internal — Confidential · Contains references to SAR-related processes (31 U.S.C. § 5318(g)(2))

Version / Date: ** **Draft v1.0 — proposal for Committee and Lead Bank approval prior to any production use

Scope. ** **This proposal defines the architecture, controls, and governance for two LLM-assisted workflows in CloudWalk’s transaction-monitoring and SAR functions. It is a design and control document, not an authorization to deploy: no workflow described here may enter production until it has passed independent validation and been approved by the Compliance Committee and the Lead Bank. Both workflows are assistive only — every SAR filing decision and every alert disposition remains a documented human judgment. All figures reference the confirmed case dataset (32,348 transactions, 501 customers, 100 merchants, 630 cards; 236 chargebacks and 350 refunds used as partial ground truth).

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1. Executive Summary

CloudWalk proposes to introduce Large Language Model (LLM) assistance into two narrowly-scoped compliance tasks: (1) drafting Suspicious Activity Report (SAR) narratives from already-validated case metadata, and (2) summarizing Tier-1 alerts to help investigators triage faster. Both tasks are labor-intensive, repetitive, and today performed manually. Neither, however, may be delegated to a model in the sense that matters to a regulator: the decision to file a SAR, and the decision to escalate or close an alert, remain human determinations made by qualified BSA/AML staff and, for filing, by the Lead Bank as the filing institution. The LLM accelerates the writing; it does not make the call.

The central architectural decision in this proposal is the choice of **extractive (constrained, source-anchored) summarization over free generative drafting**. For a document that supports a criminal referral, every asserted fact must trace to validated source data, and

examiners have consistently signalled discomfort with black-box generative systems in this setting. We therefore recommend an architecture in which the model may only compile, order, and re-phrase facts that a deterministic pipeline has already computed and validated — it cannot introduce a number, date, identifier, or allegation that is not present in its structured input.

Two further design commitments run through the whole proposal. First, personally identifiable information (PII) is minimized before the prompt: legal name, SSN, date of birth, and street address never enter the model; the case is represented to the LLM through a stable pseudonymous token, and legal identity is re-inserted only in the final document written to secure institutional storage. Second, SAR confidentiality is enforced architecturally: the drafting environment is isolated from any customer-facing system, the model provider is contractually and technically barred from retaining or training on inputs, and no path exists by which SAR content could ever reach the subject.

The remainder of this document specifies each workflow against the requested control dimensions, then sets out the cross-cutting regulatory-risk analysis, examiner-facing documentation, the minimum viable model-governance program, and a direct comparison of extractive versus generative approaches on regulatory defensibility.

2. Regulatory Framework and Supervisory Expectations

The workflows sit inside an existing, examined BSA/AML program and inherit all of its obligations. The relevant authorities and guidance are summarized below; none of them is displaced by the introduction of an LLM, and each shapes a specific control in this proposal.

Authority / Guidance	Relevance to LLM use in AML
Bank Secrecy Act (31 U.S.C. § 5311 et seq.) and 31 CFR Chapter X	Establishes the SAR obligation and program requirements. Under CloudWalk’s sponsor-bank model, the Lead Bank (FDIC member) is the filing institution (31 CFR 1020.320); CloudWalk operates as program manager and refers within agreed timelines. LLM output is an input to that process, never the filing.
SAR confidentiality — 31 U.S.C. § 5318(g)(2); 31 CFR 1020.320(e)	Prohibits disclosure — direct or indirect — that a SAR has been or may be filed. Drives the confidentiality isolation, no-retention, and anti-tipping controls in Workflow 1(g) and Workflow 2.
Model Risk Management — Revised Guidance on Model Risk Management, Federal Reserve SR 26-2 / OCC Bulletin 2026-13 (Apr. 17, 2026), which rescinded and replaced SR 11-7, OCC Bulletin 2011-12, FDIC FIL-22-2017, and the 2021 interagency	Even where an LLM tool is not a classical statistical ‘model’, examiners apply the same underlying lens carried forward from the original 2011 framework into the 2026 revision: conceptual soundness, independent validation, effective challenge, and ongoing

Authority / Guidance	Relevance to LLM use in AML
BSA/AML model-risk statement (SR 21-8 / FIL-27-2021)	monitoring, tailored proportionally to risk profile. The 2026 guidance states it is “most relevant” to institutions above \$30 billion in total assets — a threshold CloudWalk’s sponsor bank does not meet — but CloudWalk adopts the same principles voluntarily given the inherent risk profile of the products in this proposal, rather than relying on the size threshold to justify a lighter posture. Notably, the 2026 guidance explicitly excludes generative and agentic AI from its scope, with the agencies instead pursuing a separate request for information (RFI) on AI-specific model risk; no dedicated federal model-risk framework for generative AI exists as of this proposal’s date. This regulatory gap is precisely why Section 5.3 below specifies a minimum viable governance program rather than pointing to an off-the-shelf federal standard.
FFIEC BSA/AML Examination Manual	Sets examiner expectations for transaction monitoring, alert handling, SAR decisioning, and independent testing. The AI layer must be testable within this framework.
Interagency Statement on Innovative Efforts (Dec 3, 2018)	Encourages responsible innovation and pilots; a well-governed pilot that underperforms need not, by itself, draw supervisory criticism. Supports the shadow-mode / phased rollout in Section 6.
FinCEN AML/CFT National Priorities (2021)	The typologies these workflows document (fraud, TCOs, proliferation-financing / Iran nexus, corruption) map to national priorities and must be handled as legitimate, documented risk factors — not informal suspicion.
NIST AI Risk Management Framework 1.0 (2023)	Govern / Map / Measure / Manage structure used to organize the minimum viable governance program.
GLBA privacy & interagency third-party risk management (2023)	Constrain what PII may leave CloudWalk’s boundary and impose vendor due-diligence on any external model provider.

Table 2.1 — Governing authorities and the specific control each drives in this proposal.

3. Workflow 1 — SAR Narrative Auto-Draft

3.0 Architecture decision: Generative vs. Extractive

The task brief asks that two approaches be proposed and evaluated. Both are technically feasible; they differ sharply in regulatory defensibility.

Option A — Generative drafting

The LLM receives structured case metadata and writes a narrative in its own words, inferring connective tissue and phrasing freely. This is faster to build and produces fluent prose, but it is the higher-risk option: the model can assert facts, causes, or characterizations not present in the source, and those fabrications are fluent and plausible — the hardest kind to catch in review. Because generation is open-ended, it is difficult to prove to an examiner that every sentence is grounded, and difficult to reproduce a given output exactly.

Option B — Extractive / constrained compositional summarization (recommended)

The LLM may only compile, order, and reformat facts that a deterministic upstream pipeline has already computed and validated. It operates under a closed-world instruction: use only the facts provided; do not infer; if a required field is missing, say so rather than fill the gap. It cannot invent a fact not present in the input metadata. PII is anonymized before the prompt (a `customer_id` token replaces the legal name; SSN never enters the model at all), and legal-identity fields are re-inserted only in the final document exported to secure institutional storage.

Evaluation dimension	Option A — Generative	Option B — Extractive (recommended)
Speed to draft	Fastest	Fast (marginally more scaffolding)
Fluency of prose	Highest	High — sufficient for a SAR narrative
Hallucination surface	Open — can fabricate facts/causation	Bounded — no fact outside the input set
Factual traceability	Weak — assertions may be synthesized	Strong — every fact maps to a source ID
Reproducibility	Lower — output varies	High — pinned inputs reproduce output
Failure mode	Fails silently (plausible fabrication)	Fails loudly (visible [MISSING] slot)
Examiner defensibility	Invites ‘black-box’ criticism	Aligns with source-anchored, explainable expectations

Table 3.1 — Option A vs. Option B across the dimensions that matter for a filing that supports a criminal referral.

Recommendation. ** **Adopt Option B. Regulators reject black-box generative systems for criminal filings, and the extractive design is the only one under which CloudWalk can demonstrate — to its own reviewers and to examiners — that every fact in a SAR narrative traces to validated source data. All controls below (a–g) assume Option B.

3.a Architecture — how the LLM receives the case

The model never touches the raw dataset, free-text notes, or customer records. Everything flows through a deterministic pipeline and reaches the LLM as a single structured object.

- **Deterministic evidence assembly.** ** **The existing detection queries (Section 4.a of the program) and evidence-assembly routines produce, for each case, a typed Case Fact Package (CFP): a JSON object of validated fields — typology codes, transaction IDs, amounts, dates, device IDs, counts, MCC codes, tokenized merchant/customer references, and the specific thresholds crossed. Each fact carries a provenance pointer (source query and row IDs) and a validation flag.
- **Anonymization layer.** ** **Before the CFP reaches the model, direct identifiers are replaced with stable pseudonymous tokens (e.g., SUBJECT_C88888, MERCHANT_325). SSN is dropped entirely, not tokenized.
- **Prompt composition.** ** **The prompt is: (i) a system instruction fixing the role and the closed-world constraints; (ii) the CFP as the sole FACTS block; and (iii) an output template mirroring the SAR narrative structure (introduction · who/what/when/where/why/how · conclusion). Regulatory boilerplate and legal citations come from a controlled, human-maintained library — never from model generation.
- **Output & re-hydration.** ** **The model returns a draft that references only tokens and only facts present in the CFP. An automated grounding gate (3.c) checks it, a human reviews it (3.d), and only at final export are legal-identity fields re-inserted into the FinCEN narrative field and written to secure storage. The model environment never persists the re-hydrated document.
- **Deployment.** ** **A private, access-controlled endpoint (self-hosted or zero-retention contractual terms) with no provider-side training on inputs and no third-party retention of prompts or outputs.

3.b PII minimization — fields that never enter the prompt

The design follows data minimization and purpose limitation: the model receives only what the narrative genuinely requires, and identity is restored only outside the model boundary.

PII field	Enters LLM prompt?	Handling
full_name	NEVER	Replaced by a stable customer_id / subject token.
ssn_last4	NEVER	Excluded entirely — not even tokenized into the prompt.

PII field	Enters LLM prompt?	Handling
dob	NEVER	Omitted. If tenure/age-band is material, pass a derived non-identifying range.
address	NEVER	Street address excluded. If geography is a typology factor, pass only a country/region risk indicator.
email / phone	NEVER (raw)	Replaced by tokens or omitted. If a shared-contact typology matters, pass a boolean/hash-match flag, not the raw value.
customer_id / account & txn IDs	YES	Pseudonymous references only — required to anchor the evidence; carry no legal identity.

Table 3.2 — PII handling at the model boundary. Legal-identity fields are re-inserted only at secure export.

3.c Hallucination controls — ensuring generated facts match validated source data

Grounding is enforced both before generation (by constraining the input) and after (by mechanically verifying the output). The post-generation gate is the load-bearing control.

- **Closed-world instruction.** ** **The system prompt permits reference only to the FACTS block and forbids inference, added context, or outside knowledge.
- **Deterministic fact-verification gate.** ** **The draft is parsed and every atomic assertion — each amount, date, count, ID, and typology label — is diffed against the CFP. Any number, date, or identifier not present verbatim in the CFP causes that sentence to be rejected and flagged. A narrative cannot be advanced until it passes this entailment check.
- **Entity and numeric allow-listing.** ** **The only amounts, dates, and identifiers the narrative may contain are those in the CFP, verified by NER/regex extraction plus set-membership.
- **Refuse rather than invent.** ** **Missing required fields are emitted as explicit [MISSING: field] markers; any such marker blocks auto-advance and routes to a human.
- **Controlled legal language.** ** **Typology definitions, statutory citations, and required keywords (e.g., the Iran-nexus keyword IRAN-2025-A002 in SAR Field 2) are inserted from a human-maintained library, not generated.

- **Deterministic decoding.** ** **Temperature 0, fixed seed, pinned model version — narrowing output variation and supporting reproducibility (3.f).

3.d Human-in-the-loop — reviewers and mandatory sign-off

The model produces a draft and nothing more. No SAR narrative may be filed, and no SAR decision may be made, without human action; the model is never on the filing path.

Stage	Owner	Responsibility
Draft generation	Automated pipeline	Produces a constrained draft from the validated CFP.
Fact & completeness review	AML Investigator / Analyst	Verifies every fact against source; confirms the 5 W's + H; confirms no PII leakage or confidentiality issue; attests via a documented checklist.
SAR decision & sign-off	BSA/AML Officer (or delegate)	Makes the decision whether to file — a human judgment the model never makes. Signs off on the final narrative.
Filing	Sponsor Bank (Lead Bank)	Files as the filing institution once CloudWalk refers the completed, human-approved package.

Table 3.3 — Nothing auto-files. Four-eyes review is mandatory for high-risk cases (sanctions/PEP nexus, Iran keyword). Every human edit is captured as a diff and feeds the override-rate metrics in Section 5.2.

3.e Bias mitigation — equal treatment regardless of name, country, or ethnicity

Because the model works from a closed, validated fact set and asserts behavior and typology rather than character, it has little room to editorialize by identity. The controls make that structural.

- **Legitimate vs. prohibited factors.** ** **A country appearing on OFAC/FATF lists is a legitimate, documented AML risk factor and is handled as such. A subject's name, ethnicity, or a non-sanctioned nationality used as an informal suspicion driver is prohibited and is excluded from any prioritization logic.
- **Counterfactual (paired) testing.** ** **Hold the facts constant and swap the tokenized name or a non-sanctioned country; confirm the narrative's content, tone, and priority are invariant. Disparate output across synthetic cohorts is a defect to be remediated.
- **Controlled vocabulary.** ** **The model draws descriptive language from a standardized library; loaded or prejudicial descriptors are not available to it.

- **Documented review.** ** **Bias-testing results are recorded in the model card (Section 5.1) and re-run on a periodic and post-change basis.

3.f Model reproducibility — an auditable, repeatable output

- **Pinned everything.** ** **Model version/hash, decoding parameters (temperature 0, fixed seed), prompt-template version, and an immutable CFP snapshot together mean the same case reproduces the same — or fact-equivalent — output.
- **Per-SAR lineage record.** ** **For each draft, retain: CFP hash, prompt-template version, model version/params, raw model output, grounding-gate results, the human-edit diff, and the final text. Because every assertion must trace to the CFP, output cannot drift unless the CFP itself changed.
- **Retention.** ** **Lineage records are retained for at least five years, aligning with SAR supporting-documentation retention, and can be re-run to reproduce an identical audit trail on examiner demand.

3.g SAR confidentiality — never exposing SAR data to the subject (31 U.S.C. § 5318(g)(2))

- **Architectural isolation.** ** **The SAR drafting system is logically segregated from every customer-facing system, chatbot, or support tool. No shared model instance, no shared memory or vector store, and no retrieval path could surface SAR content to the subject.
- **No training, no retention.** ** **The provider is contractually and technically barred from training on, or retaining, prompts and outputs; self-hosted or zero-retention endpoints are preferred. Combined with pseudonymized inputs, there is no path for a later, unrelated query to elicit SAR facts.
- **Access control.** ** **Role-based access limits the pipeline to authorized BSA personnel, under least-privilege and full logging.
- **No indirect tipping-off.** ** **No customer-facing channel may ever generate content that reveals or implies a SAR was filed or contemplated — including veiled references to the detection system, the investigation, or the analyst. Triage summaries (Workflow 2) are internal-only and never surfaced to the customer.
- **Secure storage of identity.** ** **Re-hydrated legal identities exist only in secure institutional storage with restricted access — never in the model environment.

4. Workflow 2 — Tier-1 Alert Triage Summarization

This workflow helps an investigator orient to a Tier-1 alert faster by producing a short, consistent summary and an advisory priority. It is assistive only: the deterministic rule engine remains the system of record for tiering, and no alert can be closed or suppressed by the model.

4.a Architecture — what the model receives

The model receives structured alert fields, not raw customer PII, under the same anonymization pattern as Workflow 1.

- **Inputs:**** **alert_id, rule/typology code, tier signals (pep_flag, sanctions_match_score band, high_risk_mcc_flag, amount_usd relative to expected_avg_ticket_usd, device-sharing count, chargeback ratio), counts, dates, and tokenized entity references. No legal name, SSN, DOB, or street address.
- **Purpose:**** **orient the human analyst, not decide disposition. The deterministic engine has already assigned the SLA tier (P1/P2/P3); the model’s role is to summarize and to corroborate, flagging divergence rather than overriding.

4.b Expected output

Output element	Definition and constraints
Advisory priority score	A suggested priority (aligned to the existing P1/P2/P3 SLA bands) with a one-line rationale. Reconciled against the deterministic tier; any divergence is itself surfaced as a signal. The model’s score never lowers the deterministic tier.
3-line summary	Extractive: (1) what triggered the alert, (2) the key evidence, (3) why it may matter — drawn only from the alert fields, subject to the same grounding gate as Workflow 1.
‘Needs human review’ flag	Set TRUE by default for any sanctions/PEP nexus, any high-risk band, any ambiguous or low-confidence case, or any model-declared uncertainty. In Tier-1 all alerts receive human review; the flag raises urgency — it never gates a case out of review, and the model cannot close an alert.

Table 4.1 — The model proposes and summarizes; it never disposes. Priority is advisory and floored by the deterministic tier.

4.c Validation — measuring whether the right cases are escalated

- **Ground truth.**** **Chargeback outcomes (236), refunds (350), confirmed SAR filings, and recorded analyst dispositions.
- **Metrics.**** **Precision and recall of the high-priority / ‘needs review’ flag against eventual escalation and SAR; a confusion matrix; rank correlation between the model’s priority, the deterministic tier, and final disposition; and precision@k for the top of the priority queue. The false-negative rate — missing a case that should have escalated — is the critical metric.

- **Baseline discipline.** ** **The model must never underperform the deterministic rule engine on recall. Back-testing / look-back against historical alerts and champion-challenger comparison against the deterministic baseline are run before deployment and on an ongoing basis, feeding the monthly rule-effectiveness cycle already defined in the program's Section 4.c framework.

4.d Fallback — when the model declines or hallucinates

- **Fail to human, never drop.** ** **If the model declines, errors, times out, or returns malformed output, the alert routes to the standard manual queue at no lower than its deterministic tier.
- **Grounding failure.** ** **If the grounding gate detects an asserted fact absent from the alert data, the summary is discarded, the alert is routed to manual with a flag, and the event is logged as an incident.
- **Kill switch.** ** **The deterministic engine can run without the LLM layer; the assistive layer can be disabled at any time without stopping triage. Fallback rate is tracked as a system-health metric.

5. Cross-Cutting Requirements (Both Workflows)

5.1 Regulatory risks of using LLMs in AML

The BSA program's obligations are unchanged by the tooling; the risks below are the specific ways an LLM could undermine them, each mapped to a control already specified above.

Regulatory risk	Why it matters	Mitigating control
Hallucinated facts in a filing	A fabricated fact in a SAR undermines a criminal referral and the program's integrity.	Extractive architecture + grounding gate (3.c).
Black-box / unexplainable output	Examiners must understand why a fact was asserted or an alert escalated.	Source-anchored facts + lineage records (3.a, 3.f).
SAR confidentiality leakage / tipping-off	Disclosure that a SAR exists violates § 5318(g)(2).	Isolation, no-retention, anti-tipping controls (3.g).
PII exposure to a model endpoint	GLBA and privacy exposure if PII leaves the boundary.	Pre-prompt PII minimization (3.b).
Bias / disparate treatment	Identity-driven over/under-escalation harms effectiveness and raises UDAAP-adjacent risk.	Legitimate-factor discipline + counterfactual testing (3.e).
Automation bias	Reviewers rubber-stamping erodes the independent human judgment the BSA	Mandatory HITL, four-eyes for high-risk, override tracking (3.d, 5.2).

Regulatory risk	Why it matters	Mitigating control
Model drift / silent degradation	requires. Escalation quality can change without notice.	Ongoing monitoring vs. baseline; re-validation triggers (5.3).
Over/under-filing	Defensive over-filing buries real risk; under-filing misses it.	Precision/recall validation; recall floor vs. deterministic engine (4.c).
Uncontrolled version change	A silently updated model breaks reproducibility and validation.	Version pinning + change management + re-validation (5.3).

Table 5.1 — Regulatory risk register for LLM use, each item tied to a specific control in this proposal.

5.2 Documenting AI usage for examiners

Examiners will expect to see the AI components documented to the same standard as any other model or automated control. The artifacts below are produced and retained for each workflow.

- **Model card (per workflow).** ** **Purpose; in/out-of-scope uses; inputs (fields) and explicit exclusions (PII); model and version; decoding parameters; versioned prompts; grounding controls; HITL design; performance metrics; bias-testing results; known limitations; owners; and approval date.
- **Validation logs.** ** **Pre-deployment validation report, ongoing-monitoring dashboards, back-testing / look-back results, and drift metrics.
- **Override / edit rates.** ** **For Workflow 1, the human-edit rate and the mix of factual vs. stylistic edits — a rising factual-correction rate is a red flag. For Workflow 2, the rate at which humans change the advisory priority or disagree with the flag.
- **Decision, incident, and change logs.** ** **Per-case lineage records; incidents (grounding failures, fallbacks); and a change log recording every model/prompt version change with the re-validation that followed.
- **Data-flow and PII documentation.** ** **A data-flow / DPIA-style assessment showing what enters the model and what does not, supporting independent BSA testing coverage of the AI components.

5.3 Minimum viable model-governance program

Organized on the NIST AI RMF (Govern / Map / Measure / Manage) and consistent with the principles carried forward from SR 11-7 into the current Revised Guidance on Model Risk Management (SR 26-2 / OCC Bulletin 2026-13, Apr. 17, 2026), the minimum program has the following elements. Because the 2026 guidance excludes generative and agentic AI from its scope pending a dedicated federal framework, the program below is deliberately more prescriptive than the principles-based 2026 guidance requires of in-scope models — it is

designed to withstand examiner scrutiny in the absence of an AI-specific standard, not merely to satisfy the letter of guidance that does not yet cover this technology.

- **Governance & ownership.** **An AI/model-governance committee; a named model owner, an independent validator, and BSA/AML Officer accountability; reporting into the quarterly CloudWalk–Lead Bank governance committee; and a written policy for AI use in AML.
- **Inventory.** ***A model-inventory entry for each workflow.
- **Development standards.** ***Documented design, data governance, and PII controls.
- **Independent validation before production.** ***Conceptual soundness, outcome analysis, and effective challenge — sign-off required before any live use.
- **Human-in-the-loop by design.** ***Humans retain every filing and disposition decision.
- **Ongoing monitoring.** ***Performance, drift, override rates, and incidents against defined thresholds, with explicit re-validation triggers.
- **Change management.** ***Version pinning; any model or prompt change forces re-validation before release.
- **Third-party / vendor risk (if external model).** ***No-train and zero-retention contract terms, SOC reporting, and a documented exit plan.
- **Recordkeeping & auditability.** **At least five-year retention and full reproducibility.
- **Incident response & kill switch.** **A defined path to disable the AI layer and revert to deterministic/manual processing.
- **Training.** ***Staff training on the tools and on automation-bias awareness.
- **Independent audit.** ***Coverage of the AI components within BSA independent testing.

5.4 Extractive vs. generative — regulatory defensibility

The two approaches differ most where it counts for an examiner: whether the institution can prove that what the system produced is true to its sources.

Defensibility dimension	Generative	Extractive (recommended)
Traceability of each fact	May be synthesized; hard to source	Every fact maps to a validated source ID
Hallucination surface	Open	Bounded to the input fact set
Reproducibility / auditability	Lower	High — pinned inputs reproduce output

Defensibility dimension	Generative	Extractive (recommended)
Failure behavior	Silent (plausible fabrication)	Loud (visible missing-fact marker)
Examiner reception for a criminal filing	‘Black-box’ criticism likely	Source-anchored, explainable

Table 5.2 — For documents that support a criminal referral, extractive summarization is materially more defensible.

Bottom line. ** **Generative drafting optimizes for speed and fluency; extractive summarization optimizes for the property regulators actually require in this setting — that every asserted fact is true to a validated source and can be shown to be so. CloudWalk adopts the extractive approach for both workflows and treats the SAR narrative as a human work product that the tool assists, never authors.

6. Recommended Implementation Path

Consistent with the 2018 Interagency Statement on Innovation, deployment proceeds in controlled phases with the Compliance Committee and Lead Bank informed at each gate.

- **Design & governance sign-off.** **Finalize model cards, data-flow assessment, and the governance program; obtain Committee and Lead Bank approval to pilot.
- **Shadow mode.** ** **Run both workflows alongside current manual processes without relying on their output; measure grounding-gate pass rates, precision/recall, and override rates against the deterministic baseline.
- **Independent validation.** ** **Validate conceptual soundness and outcomes; document effective challenge; remediate before any reliance.
- **Limited production with full HITL.** ** **Enable assistive use with mandatory human review and four-eyes on high-risk cases; monitor continuously.
- **Ongoing monitoring & periodic re-validation.** **Maintain dashboards, incident and change logs, and scheduled re-validation; retain the kill switch throughout.

Assumptions & limitations. **This proposal assumes the deterministic detection and evidence-assembly pipeline is the authoritative source of facts and remains independently validated. The LLM adds no detection capability; it assists drafting and triage only. No workflow may enter production before independent validation and Committee/Lead Bank approval, and every SAR-filing and alert-disposition decision remains a documented human judgment.